



TOPIC

⇒ GPUs and Cooling Systems Theory

⇒ Basics of GPUs and cooling systems

(air vs. liquid cooling).

⇒ Identify a GPU and cooling system in a PC setup

TASKS:1

Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

RESULT

Windows

- Press `Ctrl + Shift + Esc` to open **Task Manager**.
- Click on the **Performance** tab on the left sidebar.

- Scroll down to the **GPU** section at the bottom of the list.
- The brand and model number will be displayed in the top right corner (e.g., *NVIDIA GeForce RTX 3060* or *Intel Iris Xe Graphics*).

macOS

- Click the **Apple Menu** (□) in the top-left corner of your screen.
- Select **About This Mac**.
- Look for the **Graphics** or **Chip** line to see your GPU details (e.g., *Apple M1* or *AMD Radeon Pro 5500M*).

Linux

- Open a terminal window.
- Type `lspci | grep -i vga` or `lshw -C display` and press Enter to view your graphics hardware details.

How to Identify Integrated vs. Dedicated:

- **Integrated GPU:** Built directly into the CPU processor. Examples include *Intel HD/UHD/Iris Graphics*, *AMD Radeon Graphics* (without a specific number series), or *Apple M-series chips*. They share the system's main RAM.
- **Dedicated GPU:** A separate graphics card with its own dedicated memory (VRAM). Examples include *NVIDIA GeForce/GTX/RTX series* or *AMD Radeon RX series*.

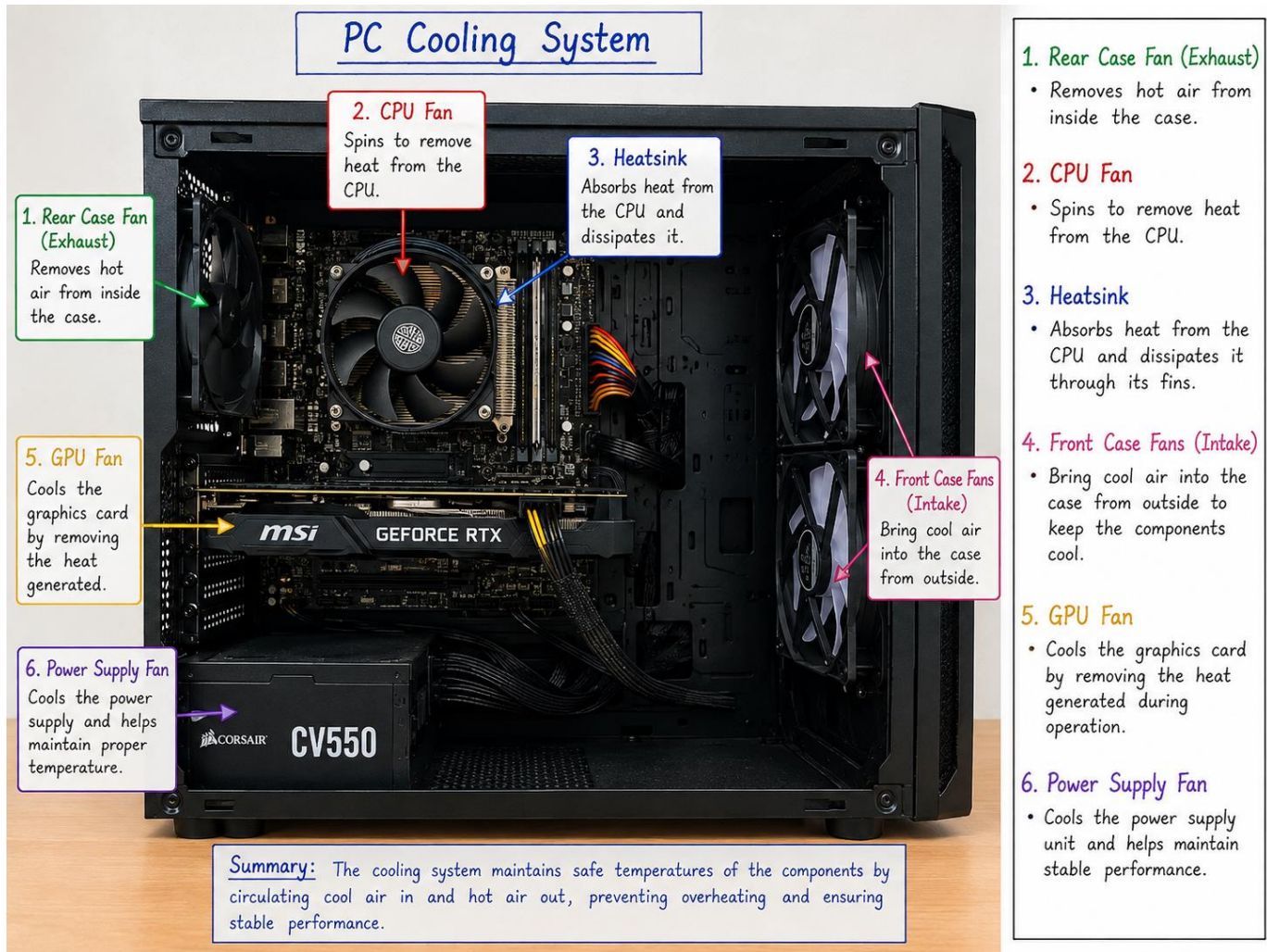
Conclusion

- **GPU Identification:** You can easily find your GPU name and brand using **Task Manager** (Windows) or **About This Mac** (macOS).
- **Integrated GPU:** Built directly into the CPU processor; ideal for everyday tasks and basic media consumption as it shares system RAM
- **Dedicated GPU:** A standalone graphics card with its own dedicated memory (VRAM); designed for heavy workloads like high-end gaming, video editing, and 3D rendering.

TASKS:2

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function

RESULT



Cooling System Components & Functions:

- **Cooling Fan:** Pushes cool air across the heatsink fins and exhausts warm air away from the CPU.
- **Heatsink:** A block made of metal fins (aluminum/copper) that absorbs heat from the CPU and spreads it across a large surface area for air cooling.

- **Heat Pipes:** Copper tubes filled with liquid/vapor that quickly transfer heat from the CPU base to the heatsink fins.

Liquid cooling setup: A PC liquid cooling system works similarly to a car's cooling engine. It transfers heat away from high-performance components (like the CPU or GPU) using liquid coolant instead of relying solely on air.

TASKS:3

Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

RESULT

Air Cooling vs liquid Cooling for GPUs

FACTOR	Air Cooling	Liquid Cooling
Cost	Generally cheaper	Generally more expensive
Cooling Efficiency	Good cooling performance for most GPUs	Usually provides better heat transfer, especially under heavy loads
Maintenance	Low maintenance; mainly cleaning dust	Higher maintenance; requires checking the cooling loop and possible leaks
Installation	Simple and easy to install	More complex to install
Noise	Fans can become noisy at high loads	Can be quieter in some setups, though pumps and radiators also make noise
Space	Usually requires less additional equipment	Requires space for a radiator, pump, and tubing

Conclusion

Air cooling is **cheaper, simpler, and easier to maintain**, while liquid cooling generally offers **better cooling performance** but costs more and is more complex.

TASKS:4

Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research

RESULT

I have chosen Liquid Cooling for my GPU.

- ➡ Simultaneously playing heavy graphics games and streaming IPL matches puts an extreme thermal load on the GPU, causing it to heat up rapidly. Liquid cooling uses coolant fluid to transfer heat away from the GPU much more efficiently, preventing overheating.

This allows the graphics card to maintain peak performance during long sessions without thermal throttling or lag. Additionally, liquid cooling operates much more quietly, preventing fan noise from bleeding into the microphone during a live stream.

Conclusion

Liquid cooling is the best choice for heavy gaming and live streaming because it prevents GPU thermal throttling, maintains peak performance during long sessions, and operates quietly to keep stream audio clear.